

Dubai Acoustic Research Laboratory
ISO Acoustic Physical & Mechanical Testing Laboratory

ACOUSTIC TEST REPORT

Customer: **RUBBER WORLD INDUSTRY LLC**
P.O. Box 2435,
Ajman, UAE

Co-sponsor: RWI INDUSTRIES (PRIVATE) LTD
RUBBER WORLD INDUSTRY (KSA)

Report Number: **169A002**

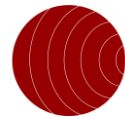
Test Standard: BS EN ISO 354:2003

Classification BS EN ISO 11654:1997
Standard: ASTM C423-22

Test Specimen: GULF.O.FLEX/GULF.O.FLEX ULTRA ELASTOMERIC INSULATION 19mm

Test dates: 25 Sep 2024





REPORT REGISTER

The following report register documents the development and issue of this and any subsequent report as undertaken by our office, in accordance with the *Quality Assurance* policy of Dubai Acoustic Research Laboratory.

Our Reference	Remarks	Issue Date
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169A002	Sound Absorption Test Report	21 st November 2024
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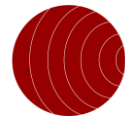
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Prepared By:	Raees Nisar Faki Acoustic Testing Engineer	18.11.2024
Reviewed By:	Paul Schwarz Laboratory Director	21.11.2024

Revision	Notes	Date
00		21.11.2024





1.0 INTRODUCTION

General

This report has been prepared for the titled project or named part thereof and should not be used in whole or part and relied upon for any other project without the written authorization of the Dubai Acoustic Research Laboratory (DARL).

DARL accepts no responsibility or liability for the consequences of this document if it is used for a purpose other than that for which it was commissioned.

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The findings expressed are relevant to the dates of the undertaken works and should not be relied upon to represent conditions at substantially later dates.

Results included therein are based on information gathered during the study. If additional information becomes available which may affect our results, DARL reserve the right to review the information, reassess any new potential concerns, and modify our results accordingly.

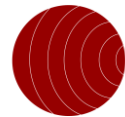
Purpose of Report

DARL has undertaken laboratory sound absorption testing of the 19mm Flexible Elastomeric Insulation at the DARL testing facilities, Jebel Ali Industrial Area 3, Dubai, United Arab Emirates.

The purpose of the sound absorption test was to determine the weighted sound absorption coefficient for the composite system with measurements made in accordance with BS EN ISO 354:2003, "*Acoustics – Measurement of sound absorption in a reverberation room*" and rated in accordance with ASTM C423-22 & BS EN ISO 11654:1997 "*Acoustics – Sound absorbers for use in buildings – Rating of sound absorption*".

Construction/mounting/installation of the test specimen was undertaken on the 25th September 2024, and measurements were undertaken on 25th September 2024 under controlled conditions.





2.0 RESULTS SUMMARY

Fully tabulated measurement data and plotted results can be found towards the end of this report. In summary, the specimen submitted for absorption testing achieved the following result:

GULF.O.FLEX/GULF.O.FLEX ULTRA ELASTOMERIC INSULATION 19mm

Absorption Coefficient (α_w) for :	0.35
NRC	0.40

* It is strongly recommended to use this single-number rating in combination with the complete sound absorption coefficient curve that can be obtained on request.

The results apply to the sample as received and is related only to the tested item and laboratory conditions as described in the report. The laboratory can make no judgment about the representativity of the tested sample. The test report ahead is valid as long as the tested constructions and/or materials are unchanged.

This single-number rating is obtained in accordance with ASTM C423-22 & BS EN ISO11654:1997 is based on a result obtained by laboratory measurements.

3.0 MEASUREMENT & METHODOLOGY CRITERIA

Measurements and calculations were undertaken in accordance with BS EN ISO 354:2003 and rated in accordance with ASTM C423-22 & BS EN ISO 11654:1997, making full allowance for current acoustic industry discussions surrounding measurement methodology and accuracy with respect to laboratory absorption testing and equipment configurations.

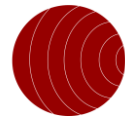
The reverberation room comprises a fully-enclosed space. The reverberant room is having the following details:

Room	Surface area	Volume	Shape	Diffusers*
Reverberation room	Total: 245.50 m² Floor: 54.30 m ² Ceiling: 54.30 m ² Walls: 136.85 m ²	249.00 m ³	Rhombus	14 pieces Total area: 28 m²

Table 1: Laboratory details.

The reverberation room is entirely enclosed and the decaying sound field in the room is sufficiently diffuse. The 19mm Flexible Elastomeric Insulation installed according to ISO-354 - mounting method 'A'. The specimen was placed on the floor in the center of the room, such that no part of the test sample is closer than 1 m to any edge of the boundary of the room.





4.0 TEST EQUIPMENT AND PROCEDURES

Measurements were undertaken with the following calibrated equipment –

Equipment	Type	DARL Ref	Serial No.	Calibration Due Date
Type 1 professional Sound Level Meter	NTi XL2 Analyzer	DARL – NTi – XL2 – S6i1	A2A-14135-E0	24-Feb-25
Microphone	Nti Audio M2211	DARL-NTiMA220-7458-S6i4	7458	24-Feb-25
Calibrator	Castle GA607	DARL-Castle GA607-S6i2	044739	24-Feb-25
Nti/Neutrik	Minirator MR Pro	DARL- MR-RAHSQ-S3i3	G2P- RAHSQ-G0	-
Loudspeaker	3x Yamaha DSR115	-	n/a	-

Table 2: Test equipment details.

Calibrations of the XL2 Type 1 sound level meter were undertaken before and after each measurement session. It was noted that no significant level of calibration drift has occurred (less than 0.1 dB).

Testing Methodology

Sound absorption testing was undertaken in accordance with ISO-354 *Acoustics – Measurement of sound absorption in a reverberation room*". The testing and data collection procedures were arranged to satisfy the following requirements:

- the number of microphone positions
- the distance between source and microphone position
- the distance between source/microphone position and the room boundaries
- the distance between two sources/microphones positions
- the distance between the microphone position and the tested specimen
- sample size
- the mounting method according to ISO-354 requirements.

The section below provides a summary of the test procedure detailed within the standard.

Microphone/source positions, tested sample location, and room shape is presented in Figure 1.



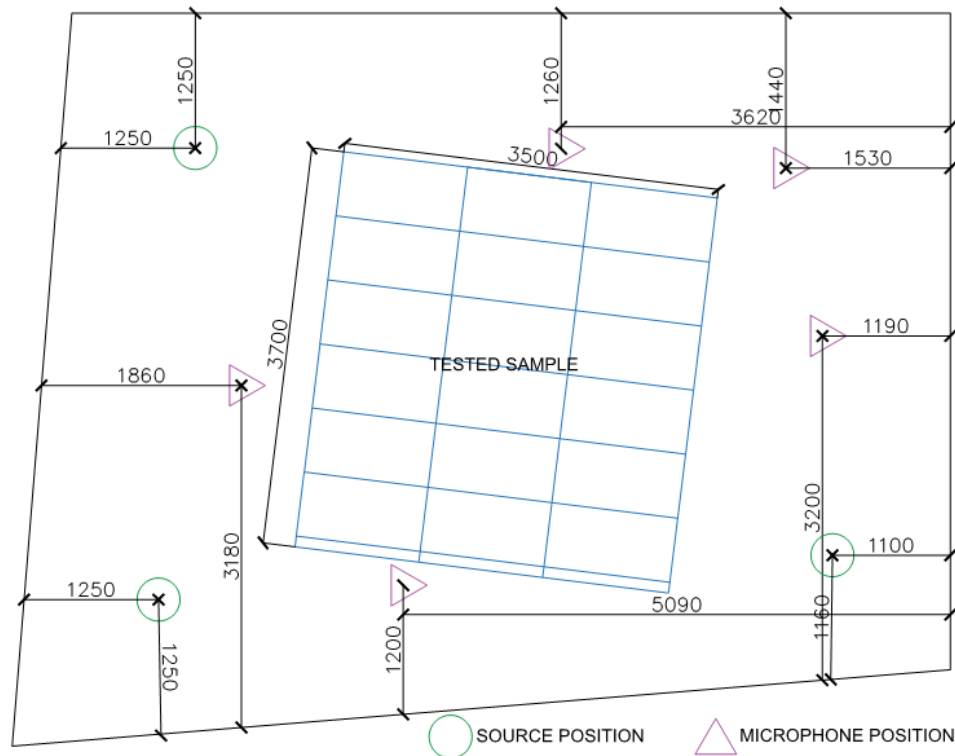
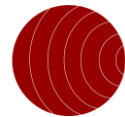


Figure 1: Reverberation room shape, microphone/source positions, and tested sample location.

Sound Absorption Testing - Summary of Test Procedure

Five microphone positions in the reverberation room are used to measure the reverberation for three nos of source positions. Measurements were undertaken with and without the tested specimen installed using an interrupted pink noise source. Ten measurements cycles are recorded for each of the fifteen loudspeaker/microphone combinations, thus representing 150 decays for each frequency band. Measurements were made in one-third octave bands with frequencies center from 100-5000 Hz.

Calculation method

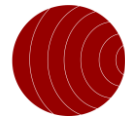
The tests were conducted in accordance with the provisions of the test method ISO 354:2003 in the Dubai Acoustic Research Laboratory reverberation room.

By means of reverberation measurements, the reverberation time of the room is measured under two conditions:

- when the reverberation room is empty
- when the construction under test is inside the reverberation room.

In general, once material is placed into the reverberation room a lower reverberation time will result. The difference in reverberation times is a measure of the amount of absorption brought into the room.

Measurements and calculations were carried out in one-third octave bandwidth from 100 to 5000 Hz. From the reverberation measurements in empty reverberation room the equivalent sound absorption area (A_1) is calculated (per frequency band) according to formula 1 and expressed in m^2 .



$$A_1 = \frac{55,3 V}{c T_1} - 4 V m_1 \quad (1)$$

in which:

V – the volume of the reverberation room [m³]

T₁ – the reverberation time in the empty reverberation room [sec.]

m₁ – "power attenuation coefficient" in the empty room, calculated according to the formula [m⁻¹]

c – propagation speed of sound in air, in m/s, calculated according to

$$m = \frac{\alpha}{10 \log(e)} \quad (2)$$

in which:

α - "attenuation coefficient" according to ISO 9613-1

In the same manner, the equivalent sound absorption area (A₂) for the room with the test specimen is calculated according to formula 3, also expressed in m²

$$A_2 = \frac{55,3 V}{c T_2} - 4 V m_2 \quad (3)$$

in which:

V – the volume of the reverberation room [m³]

T₂ – the reverberation time of the reverberation room with the test specimen placed inside [sec]

m₂ – "power attenuation coefficient" in the room with the test specimen placed inside, calculated according to a formula 2 [m⁻¹]

c – the speed of sound in the air, in m/s, calculated according to

The equivalent sound absorption area (A) of the test specimen has been calculated according to formula 4 and is expressed in m²

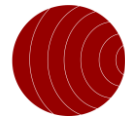
$$A = A_2 - A_1 \quad (4)$$

The sound absorption coefficient α_s of a plane absorber or a specified array of test objects shall be calculated according to formula 5:

$$\alpha = \frac{A}{S} \quad (5)$$

in which:

S - the area of the test specimen [m²]



Analysis of Results

The measurements were made in one-third octave bands. The results presented in one octave-bands are the arithmetic average of the results of the three one-third octave bands belonging to that one octave band. From those values the following one-figure ratings have been calculated and stated:

- the "weighted sound absorption coefficient a_w " according to BS EN ISO 11654:1997
- the noise reduction co-efficient "NRC" according to rated in accordance with ASTM C423-22.

5.0 TEST SPECIMEN

It has been requested by *Rubber World Industry LLC* to undertake acoustic laboratory testing to acquire the sound absorption rating for the proposed test specimens .

On the day of testing, 25th September 2024, the reverberation room conditions were measured and are presented in the table below.

Air Temperature (°C)	37
Relative Humidity (%)	50
Static Pressure (hPa)	1005.7

Table 3: Reverberation room conditions on 25 Sep 2024.

Description of the test element according to the data supplied by the client

Description	GULF.O.FLEX/GULF.O.FLEX ULTRA ELASTOMERIC INSULATION 19mm
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Table 4: Test specimen description

The dimensions of the test elements are as follows –

Width (mm)	2980
Length (mm)	3500

Table 5: Test specimen- dimensions.

During and after testing it has been noticed that the test sample has not suffered visible damage.



Drawing of the test specimen

Figure.1 -Plain view

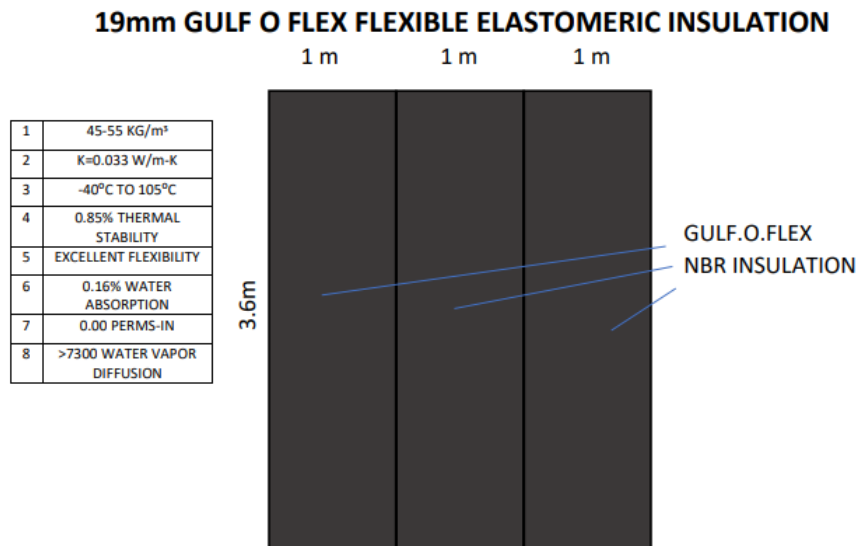
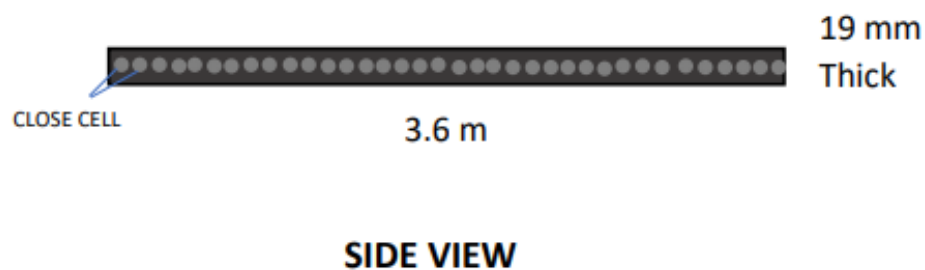
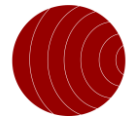


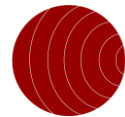
Figure.2- Side View





Pictures of the test specimen – on the days of the test





6.0 DATA

Spatially-averaged measurement data is presented below –

Frequency (Hz)	Reverberation time (s)				Frequency (Hz)
	one-third octave band		one octave band		
	T ₁ (without test sample)	T ₂ (with test sample)	T ₁ (without test sample)	T ₂ (with test sample)	
100	4.44	4.20	5.24	4.97	100
125	5.29	5.09			
160	6.01	5.62			
200	5.85	5.22	5.69	4.94	250
250	5.63	4.89			
315	5.59	4.69			
400	4.94	3.87	4.25	3.01	500
500	4.36	3.07			
630	3.45	2.09			
800	3.16	1.63	3.20	1.99	1000
1000	3.24	1.93			
1250	3.21	2.39			
1600	2.96	2.39	2.76	2.22	2000
2000	2.83	2.30			
2500	2.50	1.96			
3150	2.63	1.98	2.47	1.95	4000
4000	2.55	2.05			
5000	2.22	1.81			

Table 6: Reverberation times in the reverberation room with and without the sample.

Frequency (Hz)	A _T (m ²)	α _s (one-third octave band)	α _s (one octave band)	Frequency (Hz)
100	0.49	0.04	0.03	125
125	0.28	0.02		
160	0.45	0.04		
200	0.79	0.07	0.09	250
250	1.04	0.09		
315	1.34	0.11		
400	2.17	0.18	0.37	500
500	3.76	0.32		
630	7.36	0.62		
800	11.63	0.98	0.67	1000
1000	8.11	0.68		
1250	4.14	0.35		
1600	3.11	0.26	0.30	2000
2000	3.22	0.27		
2500	4.31	0.36		
3150	4.84	0.41	0.35	4000
4000	3.79	0.32		
5000	3.94	0.33		

Table 7: Equivalent sound absorption area and absorption coefficient

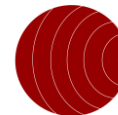
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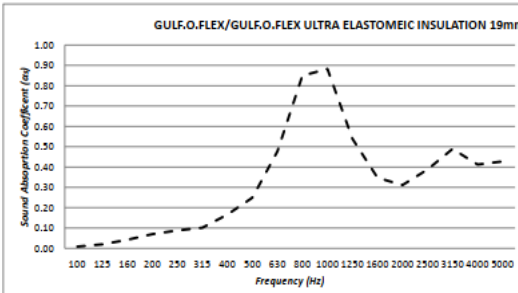


Raees Faki
Acoustic Testing Engineer
For DARL

Paul Schwarz
Laboratory Director
For DARL



CERTIFICATE

Measurement of Sound Absorption in a Reverberation Room, in accordance with ISO 354																																																												
Client:	RUBBER WORLD INDUSTRY LLC		Product Identification:																																																									
Product Manufacture	RUBBER WORLD INDUSTRY LLC		GULF.O.FLEX/GULF.O.FLEX ULTRA ELASTOMEIC INSULATION 19mm																																																									
Co-Sponsor	RWI INDUSTRIES (PRIVATE) LTD RUBBER WORLD INDUSTRY (KSA)		Date of test: 24/10/2024																																																									
Area, S_v of test element:	11.90 m^2		Test room identification: DARL																																																									
Height of the construction:	19mm ($\pm 1mm$)																																																											
Air temp. in the test rooms	34.1 °C																																																											
Relative humidity in the test room:	44 %																																																											
Static pressure:	1005.7 hPa																																																											
Receiving room volume:	249 m^3																																																											
Mounting type:	A																																																											
	<table border="1"> <thead> <tr> <th>Frequency f Hz</th> <th>α_v one-third octave band</th> <th>α_v one octave band</th> </tr> </thead> <tbody> <tr><td>100</td><td>0.01</td><td></td></tr> <tr><td>125</td><td>0.02</td><td>0.02</td></tr> <tr><td>160</td><td>0.04</td><td></td></tr> <tr><td>200</td><td>0.07</td><td></td></tr> <tr><td>250</td><td>0.09</td><td>0.09</td></tr> <tr><td>315</td><td>0.10</td><td></td></tr> <tr><td>400</td><td>0.17</td><td></td></tr> <tr><td>500</td><td>0.25</td><td>0.30</td></tr> <tr><td>630</td><td>0.48</td><td></td></tr> <tr><td>800</td><td>0.85</td><td></td></tr> <tr><td>1000</td><td>0.88</td><td>0.76</td></tr> <tr><td>1250</td><td>0.54</td><td></td></tr> <tr><td>1600</td><td>0.35</td><td></td></tr> <tr><td>2000</td><td>0.31</td><td>0.35</td></tr> <tr><td>2500</td><td>0.39</td><td></td></tr> <tr><td>3150</td><td>0.49</td><td></td></tr> <tr><td>4000</td><td>0.41</td><td>0.44</td></tr> <tr><td>5000</td><td>0.43</td><td></td></tr> </tbody> </table>	Frequency f Hz	α_v one-third octave band	α_v one octave band	100	0.01		125	0.02	0.02	160	0.04		200	0.07		250	0.09	0.09	315	0.10		400	0.17		500	0.25	0.30	630	0.48		800	0.85		1000	0.88	0.76	1250	0.54		1600	0.35		2000	0.31	0.35	2500	0.39		3150	0.49		4000	0.41	0.44	5000	0.43			
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Rating in accordance with the following standards: NRC (ASTM C423) 0.40 α_w (ISO 11654) 0.35 Absorption class: D																																																												
No. of test report : 169A002 Date: 24/10/2024		Name of test Institute : ISO Acoustic P&MTL Signature : 																																																										

